LPM 2026 ROME — LIVE PERFORMERS MEETING

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1959

An audio-visual installation · Federico Corradi

In 1959, David Hubel and Torsten Wiesel lowered an electrode into the visual cortex of a cat and moved a bar of light across its field of view. A single neuron answered, and its spikes were audible as clicks on tape. They received the 1981 Nobel Prize for what those clicks meant: that seeing is built, cell by cell, from single features of the world.

This installation plays that recording into a mixed-signal chip I designed and fabricated, containing sixteen analog spiking neurons. Each neuron has a membrane, a leak and a threshold. It has no clock and computes no numbers: it answers in continuous time, in physics. It does not answer identically. No two neurons, born or made, ever do.

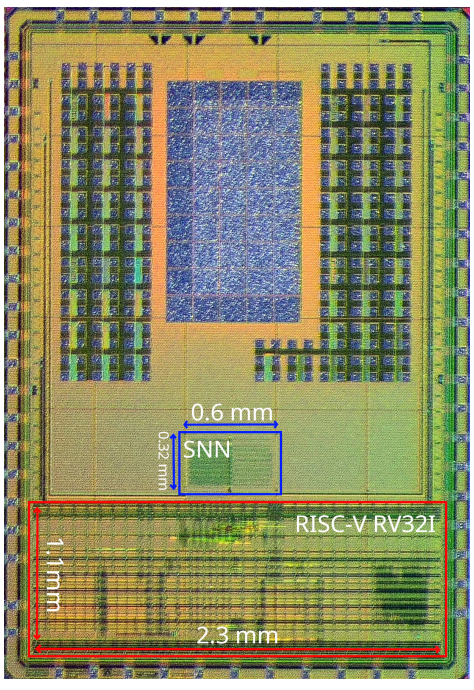
The visitor hears both: a cat's cortex from 1959 and a silicon analog neuron from 2026, made with state-of-the-art microelectronics, in the same room, sixty-seven years apart, almost in unison. The chip is exposed on a plinth beside a print of the neuron and synapse layout.

Those same clicks are where machine vision began. Today's neural networks are a diagram of this cat. What they copied was the wiring. What they left behind was the body.

Sixteen neurons is not a brain. It is the start of a different bet: that machines which have to perceive, in real time, in the world, on the power the world can spare, should be made of physics rather than numbers. How far that goes, nobody knows yet. In 1959, nobody knew either.



The cat, 1959. Electrode in the visual cortex, spikes on a loudspeaker. *Still: "Hubel & Wiesel 2: Cat Experiment".*



The chip under a microscope. A small computer and sixteen artificial neurons on one piece of silicon just **3.6 × 5.2 mm**. Like the cat's, these neurons run on current, not numbers. What you hear is one neuron's voltage, amplified straight from the chip — no computer in between.

What they copied was the wiring. What they left behind was the body.

LISTEN

Put on the headphones. One ear carries the 1959 recording; the other carries the silicon neuron answering it. Stay as long as you like — the piece never repeats itself the same way twice.

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